Assignment 1

Recommended readings

- Lecture Slides as starting literature
- https://developers.google.com/web/tools/chrome-devtools/#discover
- http://httpd.apache.org/docs/2.4/getting-started.html
- https://developer.mozilla.org/en-US/docs/Web/HTML/Attributes
- https://developer.mozilla.org/en-US/docs/Web/HTML/Element/footer
- https://developer.mozilla.org/en-US/docs/Web/HTML/Element/a
- MDN Links within slides for details

Exercise 1 – HTTP Requests 0,1 Points

a) Use the *Developer Tools* of your web browser (e.g., Chrome) to inspect the HTTP requests when opening the website of www.aau.at and answer the following questions:
   - How many requests are sent, how much data is transferred, and how long does it take to load the site from the server?
   - What is the *response status code* of the first request?
   - What type of web server is serving this website?
   - What types of resources are requested?
   - Which request is the slowest?
   - Which cookies does the site store?

b) Repeat the procedure and answer the same questions for www.orf.at

c) Open up www.google.at, clear the content of the network inspection window, and enter the word "Klagenfurt" into the search field. While entering the word, look at the network requests and explain what happens.

d) What is the cache? How can you delete it? When could it be useful to disable or delete the cache?

Exercise 2 – Setting up Apache 0,1 Points

Install Apache HTTP Server version 2.4 and open http://localhost in your browser. You should see the text "It works".

Additionally, answer the following questions:

- Describe the purpose of `conf/httpd.conf` and explain the meaning of the following terms: ServerRoot, DocumentRoot, Virtual Host
- What is the loopback IP address `127.0.0.1` and the hostname `localhost` used for?

- What is the purpose of the `index.html` document? Find the document, edit its content, and open http://localhost again. Create a folder "mytest" in the documents root folder and add some random files and folders to the new folder. Open http://localhost/mytest in your browser and explain the result.

Additional Info:

- **For MS Windows** you can obtain binaries for apache from https://www.apachelounge.com/download/. Extract the contents of the download to c:\programme\apache. For starting the server execute the file c:\programme\apache\Apache2.4\bin\httpd.exe
- **For Linux:** Use your favorite package manager and install apache httpd 2.4.
- **For Mac:** Have a look at: https://medium.com/@aaron_lu1/apache-use-macos-builtin-apache-1f17beeb3f0a

Exercise 3 – HTML Page 1 0,1 Points

Study the different HTML concepts (elements, tags, and attributes).
Write a simple web page that looks as follows using HTML only (**NO CSS!**): Test this page with your Apache setup by placing it in the DocumentRoot folder (see previous exercise). Make sure to use appropriate meta tags in the header and give your page a title!

This is a top-level heading

After the heading, a horizontal line follows.

The first paragraph loads the AAU logo directly from **https://www.aau.at/**, with a square dimension of 150x55 pixels:



This division contains an unordered list with nested ordered lists:

- First item
    1. Property 1
    2. Property 2
- Second item
    1. Property 1
    2. Property 2
- Third item
    1. Property 1
    2. Property 2

This inline-span contains a link to the website of *Universität Klagenfurt*, opening in a new window: Click here
Finally, this document contains a horizonal line.

Exercise 4 – HTML Page 2 0,1 Points

a) Create the following HTML files:

- `index.html`
- `img1.html`
- `img2.html`
- `img3.html`

The folder `ex1-4`, contains three image files `img1.jpg`, `img2.jpg`, `img3.jpg`, and the corresponding thumbnails `img1-thumb.jpg`, `img2-thumb.jpg`, `img3thumb.jpg`, and text files containing an image title and a short description of each image.

The `index.html` page should contain the heading “Image Gallery” and then show the thumbnails of each image as a section with the ID "gallery-section". When hovering over an image with the mouse, the image title should be shown. When clicking on an image, the corresponding page (I.e., `img1.html` for `img1.jpg`) should be opened.

The details pages for each image should contain the image title as title and as heading. Then, the image should be shown. After the image, the textual descriptions should be displayed. When hovering over the image the text “Open image [imagetitle] in full” should be shown. When clicking on the image, the image jpeg file should be opened in a new tab.

At the bottom of each image page there should be a link to the next and to the previous image page. In addition, there should always be a link to “home” navigating to the index page.

Make sure to use the `alt` attribute in a meaningful way. Why is this attribute and a meaningful value for it important?

Put all pages under the document root of your apache installation and test the web page.

b) Change `img1.html` to `img3.html` in such a way that after 7 seconds, the next picture page is loaded. After the last image, the first one should be loaded.

Exercise 5 – HTML Tables 0,1 Points

Open the file *ex5.html* and review the existing code. First, focus on the HTML code in the `body` tag and sketch the expected page on a piece of paper. Then, focus on the CSS rules in the `style` tag and think about their effects.

Complete the table with the data from below. Please note that no extra CSS rules need to be created or manipulated. A correct HTML structure and the correct use of the scope attribute automatically lead to the correct styling of the table.

Top 5 universities in the world

University	Country	Ranking
University of Oxford	United Kingdom	98.5
Massachusetts Institute of Technology	United States	98.1
Harvard University	United States	97.7
Princeton University	United States	97.5
University of Cambridge	United Kingdom	97.4

University rankings 2025

Note: Also add the link to each university's webpage in the table.
Link to the rankings of universities (required for the caption):
https://www.timeshighereducation.com/world-university-rankings/2025/world-ranking

Exercise 6 – Forms 0,1 Points

Open the file *ex6.html* and extend the HTML form for a car rental agency (details follow). Do not use CSS styles for now. Be sure to apply labels and field sets properly. Select appropriate HTML elements for each requirement and only allow valid values as input. Use HTML checks (e.g., specific input types or pattern attributes) and no JavaScript. Provide example data for the user (placeholder) for each text field. Create an HTML form for entering contact data, rental specifications, and final comments.

A car rental agency allows customers to rent vehicles based on their contact information, rental details, and preferences. The form should gather customer contact details (title, name, email, and phone number). Rental details include the pickup date, return date, and pickup location. Customers should be able to specify the number of passengers as well as part of the rental details. Finally, customers can provide vehicle preferences such as vehicle color and the type of vehicle they prefer (Sports car, Combi, Truck, Van, SUV, or Luxury). The final comments should allow the customer to enter no more than 512 characters of free text. You should use different types of input for title (select), color (color picker), type (radio), dates (date picker), and passengers (number input).

When you submit the form, the data is transmitted via email (if you have an email client configured). Ensure that the data in the email is useful for the recipient. E.g., for each value, there must be a key identifying the type of data. All fields except the comments are required. There should also be a "reset" button for clearing all currently entered data.

Hints: Use Google Chrome or Firefox for submitting the form. Submitting the form via email is possible by combining the form's `action` attribute and `mailto`-links. To specify useful names, consider using `name` attributes.

Exercise 7 - CSS Selectors 0,1 Points

Consider the HTML Structure given in file *ex7.html*. Explain what happens if you apply separately each of the following style definitions to the HTML structure. Uncomment one rule at a time in file *ex7.html* and explain the result:

```
#content, p { color: red; }
#content p { color: red; }
#content h2 p { color: red; }
#content > p { color: red; }
#content h1 p { color: red; }
div p:nth-child(2) { color: red; }
div p:nth-child(2n) { color: red; }
h2 {color: red;}
div.othercontent > p:first-child { color:red; }
div.othercontent h1 + p  { color:red; }
div.othercontent h1 ~ p { color:red; }
a[title^=other]:link { color:red; }
body a[title^=other]:link { color:red; }
.othercontent * p { color: red; }
```

Note: You can use the developer console of your web browser to apply the given CSS rules.

Exercise 8 - CSS Layout 0,1 Points

Create CSS rules for the file *ex8.html* and include them in the style element. You must not change any other content of the file. The page should be rendered as shown in the figure on the next page, according to the following guidelines:

- The text “CSS in action” in the header should be formatted in the font verdana with a size of 2em and a margin of 10px;
- The background color of the header should be red 187, green 204, blue 232.
- The logo with the id “logoImage” should be positioned on the right of the text “CSS in Action” and should have a margin of 10px and a height of 4em.
- The menu (Topic 1 to Topic 4) should be rendered in one line (see example) and with the background color red 20, green 90, blue 204. There should be a horizontal spacing of 20px between every item.
- The second menu item should be rendered in red.
- The heading of the second topic should also be rendered in red.
- All h1 headings directly nested within sections that are directly nested within the content div should be rendered in the font Verdana and a size of 1.5 em.
- The footer should be rendered in the same background color as the menu. The list items within the footer should be displayed in one row and centered.
- The footer should be at the bottom of the browser window, no matter how large it is.
- External links (for the sake of simplicity: links starting with HTTP) appearing anywhere in the content should get a disclaimer “(external)”.
- In every second section, the image(s) should be aligned to the right, otherwise to the left. Images should have a width of 100px.

The use of **flexbox** is strongly recommended. Other solutions, however, are also possible and accepted.
To get an overview of flexbox rules and features, check out the following page:
https://css-tricks.com/snippets/css/a-guide-to-flexbox/

Note: You will need to apply paddings and margins for some elements. Use values approximating the layout of the figure below. For easier CSS debugging, use the developer console of your browser.

CSS in Action

Topic 1 Topic 2 Topic 3 Topic 4

Topic 1



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Topic 2

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Topic 3



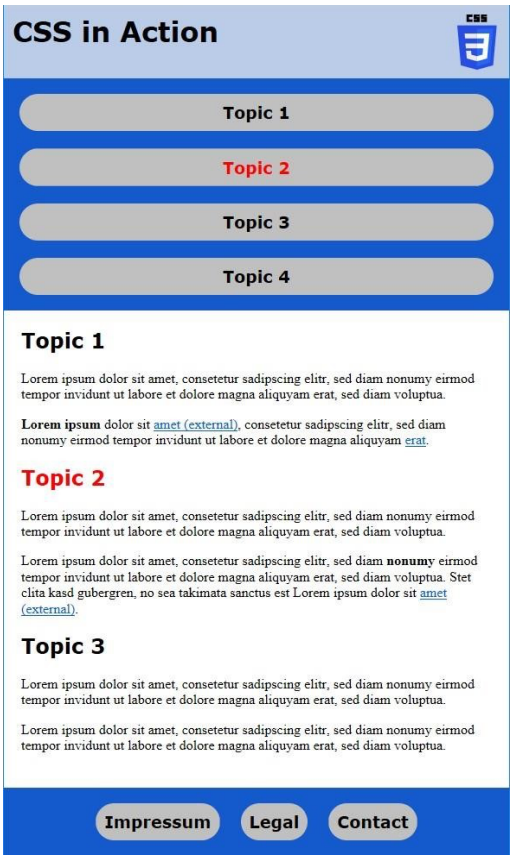
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Impressum Legal Contact

Exercise 9 – Responsive Layout 0,2 Points

Change your layout of the document from Exercise 8 so that it is rendered as follows in a browser pane with less than 600 pixels:



Note: You can easily test your solution by resizing your browser window to below 600 pixels in width. Alternatively, you can use the developer tools of your browser to specify specific device characteristics.

Recommended Project Preparations

Consider everything you learned in this assignment and apply it to plan the layout and design of your project.